

Cryo-EM structure of the CBC-ALYREF complex

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Abstract

In eukaryotes, RNAs transcribed by RNA Pol II are modified at the 5' end with a 7-methylguanosine (m⁷G) cap, which is recognized by the nuclear cap binding complex (CBC). The CBC plays multiple important roles in mRNA metabolism including transcription, splicing, polyadenylation and export. It promotes mRNA export through direct interaction with ALYREF, which in turn links the TRanscription and EXport (TREX) complex to the 5' end of mRNA. However, the molecular mechanism for CBC mediated recruitment of the mRNA export machinery is not well understood. Here, we present the first structure of the CBC in complex with a mRNA export factor, ALYREF. The cryo-EM structure of CBC-ALYREF reveals that the RRM domain of ALYREF makes direct contacts with both the NCBP1 and NCBP2 subunits of the CBC. Comparison of CBC-ALYREF to other CBC and ALYREF containing cellular complexes provides insights into the coordinated events during mRNA transcription, splicing, and export.

eLife assessment

This **important** study reports the cryo-electron microscopy structure of a multi-protein complex that recognizes the 5'-end cap of mRNAs and plays a critical role in mRNA export. The structural analyses and biochemical assays in this study provide **convincing** evidence to support the major claims of the authors, although the inclusion of more functional characterizations in cell-based systems would have strengthened the study. This paper would be of interest to structural biologists and RNA biologists working on mRNA metabolism.

Introduction

The nuclear cap binding complex (CBC) binds to the m⁷G cap of RNAs transcribed by RNA pol II. It is comprised of NCBP1 (also known as CBP80) and NCBP2 (also known as CBP20). This heterodimeric CBC can form a variety of interactions with different proteins to promote mRNA processing and influence the fate of the transcript. As such, the CBC regulates gene expression at multiple levels, ranging from transcription and splicing to nuclear export and translation^{1,2,3}.

The CBC is one of a myriad of protein factors that associate with newly synthesized transcripts. These factors package mRNAs into compacted ribonucleoprotein particles (mRNPs). While the overall structural arrangement of mRNPs is not known, factors such as serine/arginine-rich (SR) proteins are suggested to contribute to mRNP compaction. Prior to export, mRNPs acquire the export receptor NXF1-NXT1 to gain access to the nuclear pore complex (NPC)³. NXF1-NXT1 can interact with the FG repeats of the nucleoporin proteins in the NPC to mediate mRNA export^{4,5}. Intriguingly, early electron microscopic studies on the Balbiani ring mRNPs in *C. tentans* showed that mRNPs translocate through the NPC in a 5' to 3' direction⁶. Recent single molecule work on human mRNPs also suggests mRNA is exported 5' end first⁷.

The CBC promotes nuclear mRNA export through its interaction with a key mRNA export factor ALYREF. ALYREF in turn binds to DDX39B (also known as UAP56) and by extension, the entire TRanscription-EXport (TREX) complex at the 5' end of the RNA⁸. The TREX complex is conserved from yeast to humans^{9–13}. The human TREX complex is composed of THOC1, 2, 3, 5, 6, 7, and the DEAD-box helicase DDX39B¹⁴. Interestingly, ALYREF was also identified as the THOC4 subunit of the TREX complex, indicating that the function of ALYREF is tightly integrated with the TREX complex. ALYREF and its yeast ortholog Yra1 contain UBMs (UAP56-binding motifs) that mediate the interaction with DDX39B and Sub2, respectively^{10,15–17}. The yeast CBC was also shown to facilitate the recruitment of Yra1 onto nascent RNA¹⁸. ALYREF and TREX play central roles in mRNA export through direct interactions with various factors including the export receptor NXF1-NXT1^{19–21}. Their association with the 5' cap of mRNAs in particular is a key step in the export process. However, thus far the molecular mechanism underlying how ALYREF bridges the CBC and TREX remains unclear.

ALYREF and the associated TREX complex are not only required for cellular mRNA export but also can be hijacked for nuclear export of some viral mRNAs, such as those of herpes viruses. Two well studied examples are Herpes simplex virus (HSV-1) ICP27 and Herpesvirus saimiri (HVS) ORF57^{22,23}. Both HSV-1 ICP27 and HVS ORF57 directly target the host ALYREF protein^{24,25}. Structural studies show that they recognize overlapping surfaces on the RRM domain of ALYREF. How these viral factors affect host CBC-ALYREF interaction and function is not known.

The highly integrated nuclear mRNA processing and mRNP packaging also require the actions of multi-functional splicing factors, such as the SR protein SRSF1 and the exon junction complex (EJC). SRSF1 couples transcription, splicing, and export through direct interactions with the CBC, spliceosome, NXF1-NXT1, and RNA^{26–28}. Of note, ALYREF also has binding activities to these factors. Therefore, the functions of ALYREF and SRSF1 in mRNA processing and export are likely interconnected. In higher eukaryotes, the EJC is deposited 20–24 nucleotides upstream of the exon junctions during splicing^{29–31}. The EJC was initially shown to associate with ALYREF through a WxHD motif in the N-terminal unstructured region of ALYREF³². More recently, multiple binding interfaces were shown between ALYREF and the EJC¹². How the CBC-ALYREF connection affects the function of EJC-ALYREF remains elusive.

To better understand the multiple functions of ALYREF and the CBC in RNA metabolism, we carried out structural and biochemical studies. We present cryo-EM structures of the CBC and CBC-ALYREF complexes at 3.51 Å and 3.16 Å resolution, respectively. The CBC-ALYREF structure reveals that both the NCBP1 and NCBP2 subunits of the CBC interact with ALYREF. Conformational changes in the CBC are observed that accommodate the interaction. Both HSV-1 ICP27 and HVS ORF57 target the binding interface between the RRM domain of ALYREF and the CBC. We suggest that these viruses not only hijack host pathways to export their own RNA but could also inhibit host RNA metabolism through their interactions with ALYREF. Structural overlay of CBC-ALYREF and EJC-ALYREF reveals that both the CBC and the EJC bind to the RRM domain of ALYREF in a mutually exclusive manner. This suggests that ALYREF's interaction with the EJC is favored after ALYREF dissociates from the CBC, or as an independent event.

ALYREF directly interacts with CBC

Recombinant human ALYREF protein was shown to interact with the CBC in RNase-treated nuclear extracts⁸. Here we used purified recombinant proteins to further investigate the molecular interactions between ALYREF and the CBC (**Figure 1**). It is well known that RS domain containing proteins, including ALYREF, exhibit low solubility and are prone to aggregation. The addition of glutamic acid and arginine to the buffer can increase protein solubility and stability³³. Indeed, with a 1:1 mixture of glutamic acid and arginine, we were able to purify a GST-tagged ALYREF (residues 1-183) protein containing the N-terminal region and the RRM domain of ALYREF (**Figure 1A**). This construct includes the WxHD motif (residues 87-90), whose mutation was shown to affect interaction with the CBC in an immunoprecipitation study³². GST pull down assays showed that ALYREF directly interacts with the CBC (**Figure 1B**).

To aid in structural studies, we tested solubility of several ALYREF orthologs and found that mouse ALYREF2 (mALYREF2, residues 1-155) exhibited better solubility than the ALYREF construct utilized above. The conserved UBM motifs, the WxHD motif, and the RRM domain are nearly identical between ALYREF and mALYREF2 (**Figure 1-figure supplement 1**). Indeed, like ALYREF, mALYREF2 directly interacts with the CBC (**Figure 1B**). Since the ALYREF/mALYREF2 interaction with the CBC is conserved and mALYREF2 exhibits better solubility, we focused on mALYREF2 in the cryo-EM investigations.

Cryo-EM structures of CBC and CBC-ALYREF complexes

We first determined the cryo-EM structure of the human CBC complex at 3.51 Å (**Figure 2A**, **Figure 2-figure supplement 1** and **2**, **Table 1**). The cap analog m⁷GpppG was added in our cryo-EM study and the electron density of the cap moiety is clearly visible bound to NCBP2. Compared to unliganded CBC, the cap analog induces significant rearrangements in both the N-terminal extension and the C-terminal tail of NCBP2 (**Figure 2B**) to form critical interactions. For example, the N-terminal extension (residues 16-28) swings toward the central globular domain of NCBP2 and positions the Y20 residue to sandwich the cap analog with Y43 of NCBP2. These conformational changes were also observed in the crystal structures of ligand bound CBC^{34,35}. Overall, the cryo-EM structure of the CBC determined here resembles the previously reported crystal structures of the liganded CBC, with root mean squared deviation (RMSD) of 0.89 Å and 0.87 Å for NCBP1 and NCBP2, respectively³⁴.

We next determined the cryo-EM structure of the CBC-mALYREF2 complex in the presence of the cap analog m⁷GpppG at 3.16 Å resolution. The structure shows that the RRM domain of mALYREF2 binds to both NCBP1 and NCBP2 subunits (**Figure 2C** and **2D**). The N terminal region of mALYREF2 (residues 1 to 73) does not show traceable density and is possibly disordered. Compared to the CBC alone cryo-EM structure, conformational changes are observed at the mALYREF2-CBC interface. Notably, a loop formed by residues 38-45 of NCBP1 becomes ordered and moves toward mALYREF2 (**Figure 2D**).

CBC-ALYREF interfaces

The interfaces between mALYREF2 and the CBC involve 19, 14, and 5 residues of mALYREF2, NCBP1, and NCBP2, respectively (**Figure 3**). The RRM domain of mALYREF2 assumes a canonical $\beta_1\alpha_1\beta_2\beta_3\alpha_2\beta_4$ topology (**Figure 3A**), forming an α -helical surface and a β -sheet surface. The α -helical surface recognizes the CBC through extensive hydrophilic and hydrophobic interactions (**Figure 3B**). The α_1 helix of mALYREF2 is enriched with acidic residues and makes key hydrophilic interactions with NCBP1. For example, E97 forms salt bridges with K330 and K381 of NCBP1. Y135 on the α_2 helix of mALYREF2 makes a hydrogen bond with K330 of NCBP1. The importance of this interface between ALYREF and NCBP1 is highlighted by a K330N mutation

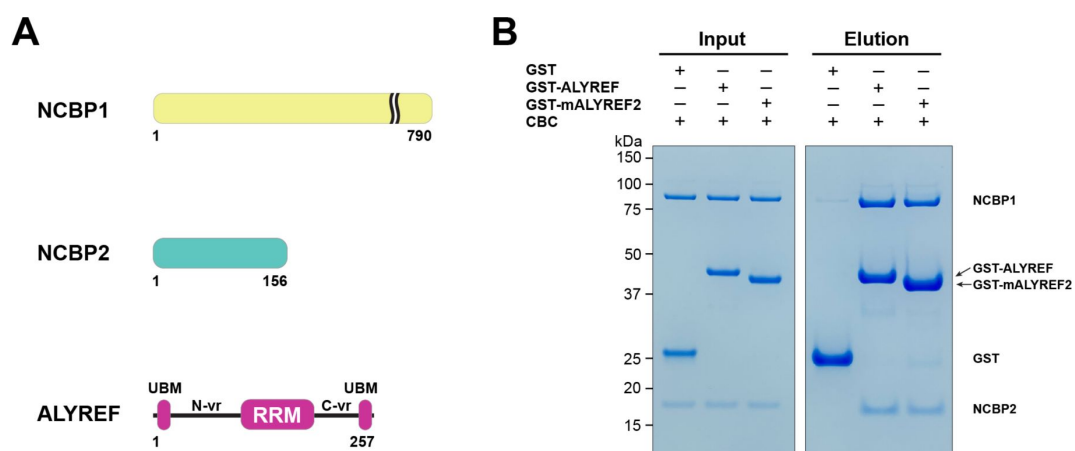


Figure 1.

ALYREF directly binds to CBC.

(A) Schematics of the NCBP1 and NCBP2 subunits of the human CBC and ALYREF. ALYREF contains a central RRM domain. The RRM domain is connected to two conserved motifs (UBMs) at both termini through variable regions (N-vr and C-vr). (B) ALYREF directly interacts with the CBC. In vitro GST pull down assays were performed with purified recombinant human CBC and GST-tagged ALYREF (residues 1-183) or mALYREF2 (residues 1-155). Results are representative of three independent experiments.

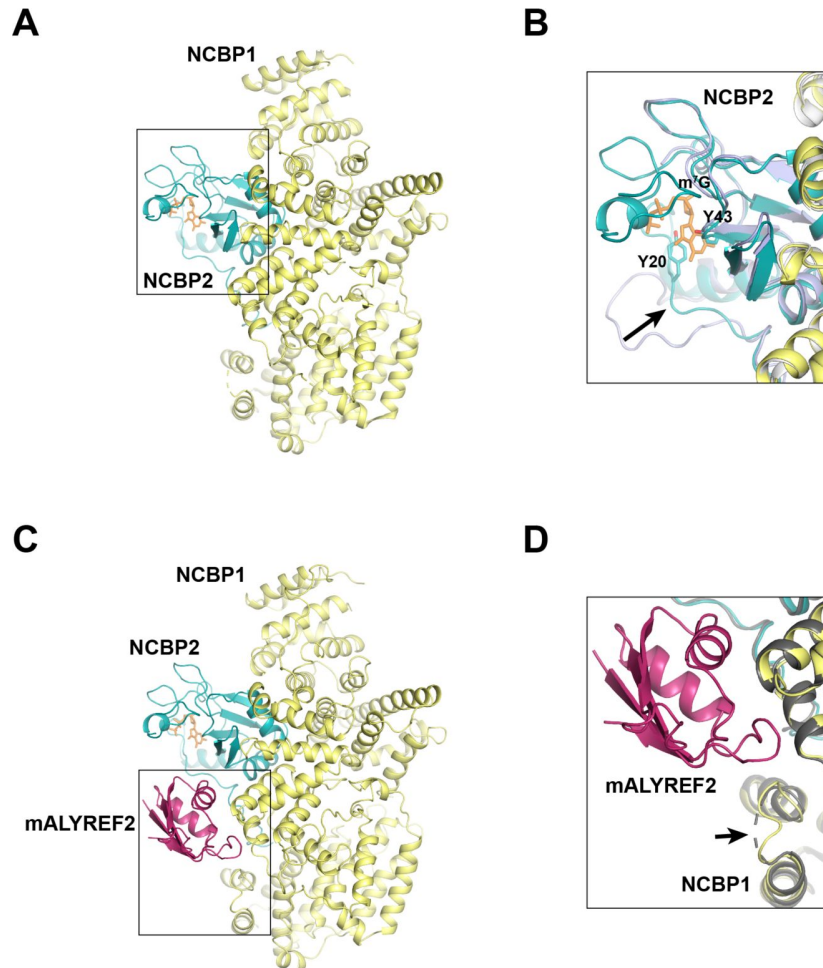


Figure 2.

Cryo-EM structures of CBC and CBC-mALYREF2

(A) Overall architecture of the CBC complex. NCBP1 and NCBP2 are colored in yellow and teal, respectively. The cap analog is shown as orange sticks. (B) Comparison of the CBC cryo-EM structure to the unliganded CBC crystal structure (PDB ID 1N54). The cryo-EM structure is colored as in (A). Unliganded CBC crystal structure is colored in light blue. Arrow indicates the conformational change in the N-terminal extension of NCBP2 upon cap analog binding. (C) Overall architecture of the CBC-mALYREF2 complex. NCBP1, NCBP2, and mALYREF2 are colored in yellow, teal, and red, respectively. The RRM domain of mALYREF2 binds to the CBC. (D) Comparison of the CBC-mALYREF2 and the CBC cryo-EM structures. CBC structure is colored in gray. Arrow indicates conformational changes in a loop (residues 38-45) of NCBP1 upon mALYREF2 binding.

	CBC-mALYREF2 (EMDB EMD-####) (PDB ####)	CBC (EMDB EMD-####) (PDB ####)
Data collection and processing		
Microscope/Camera	Glacios/Falcon 4i	
Voltage (kV)	200	
Electron exposure (e-/Å ²)	52	
Defocus range (μm)	1.0 to 2.0	
Pixel size (Å)	0.732	
Symmetry imposed	C1	C1
Final particle images (no.)	439,849	142,972
Resolution at 0.143 FSC (unmasked, Å)	3.8	4.1
Resolution at 0.143 FSC (masked, Å)	3.16	3.51
Refinement		
Model composition		
Protein residues	974	880
Ligands	1	1
B factors (Å ²)		
Protein	208.5	229.1
Ligand	215.8	239.8
R.m.s. deviations		
Bond lengths (Å)	0.002	0.002
Bond angles (°)	0.436	0.464
Validation		
MolProbity score	1.59	1.64
Clashscore	10.09	11.29
Poor rotamers (%)	0	0
Ramachandran plot		
Favored (%)	97.7	97.7
Allowed (%)	2.3	2.3
Disallowed (%)	0	0

Table 1.

Cryo-EM data collection, refinement and validation statistics

found in human uterine corpus endometrial carcinoma^{36–38}. In addition, the loop between $\alpha 2$ and $\beta 4$ of mALYREF2 forms hydrophobic interactions with NCBP1. V138, P139, and L140 of mALYREF2 bind to a hydrophobic pocket on NCBP1 formed by A334, V337, and L382. Furthermore, at the NCBP1-mALYREF2 interface, the conformational change of the NCBP1 loop (residues 38–45), observed when compared to the CBC alone structure, enables K41 of NCBP1 to interact with mALYREF2 (**Figure 2D** and **3B**).

The interface between mALYREF2 and NCBP2 is near the m⁷G binding pocket (**Figure 3C**). The $\alpha 2$ helix of mALYREF2 contacts S13 and Y14 in the N-terminal extension of NCBP2. S13 and Y14 also directly interact with NCBP1 and are thought to enable the hinged motion of the N-terminal extension (residues 16–28) upon binding to the cap³⁵. In addition, the $\alpha 1$ helix of mALYREF2 is in proximity to the R105 and I110 residues of NCBP2. Mutations of these interacting residues are associated with human cancers. For example, the NCBP2 R105C mutation has been reported in colorectal cancer and the NCBP2 I110M mutation has been found in head and neck cancer^{36–38}. NCBP2 exhibits a positively charged groove extending from the cap binding site which is suggested to be an RNA-binding site³⁴. Upon mALYREF2 binding, this groove is buried. Interestingly, mALYREF2 features a positively charged surface near the m⁷G site (**Figure 3D**). Conceivably, this positively charged surface on mALYREF2 could serve as an RNA binding site for the nucleotides following the cap.

Based on the CBC-mALYREF2 structure, we generated mutations (mut-1 and mut-2) on the RRM domain of human ALYREF (ALYREF-RRM, residues 103–183) to validate its interaction with the CBC. For ALYREF-RRM-mut-1 (Y166R/V169R/P170R), mutated residues are localized on the $\alpha 2$ - $\beta 4$ loop and correspond to residues Y135/V138/P139 in mALYREF2 (**Figure 3A, B**). For ALYREF-RRM-mut-2 (E124R/E128R), mutated residues are localized on the $\alpha 1$ helix and correspond to residues E93/E97 in mALYREF2 (**Figure 3A, B**). In agreement with the CBC-ALYREF structure, we found that the RRM domain of ALYREF directly interacts with the CBC, albeit with weaker interaction compared to ALYREF (residues 1–183) (**Figure 1B** and **4A**). The difference likely results from the WxHD motif (residues 87–90) localized in the N-terminal region of ALYREF. Evidence suggests that mutation of the WxHD motif reduces ALYREF's interaction with the CBC³². The WxHD motif may represent a second binding site for the CBC that remains to be characterized. Importantly, compared to the wild type protein, both ALYREF-mut-1 and mut-2 show reduced binding to the CBC (**Figure 4A**). Together, the mutagenesis studies validate the CBC-ALYREF interfaces observed in the structure.

The CBC-mALYREF2 structure reveals that the interaction between ALYREF and the CBC mainly involves the NCBP1 subunit (**Figure 3B** and **3C**). We further dissected the interaction between ALYREF and individual NCBP1 and NCBP2 subunits using GST pull down assays. NCBP1 can be efficiently pulled down by GST-ALYREF, whereas NCBP2 did not show detectable interaction (**Figure 4B**). These results are consistent with the structural observations and indicate that NCBP1 is the major subunit of the CBC to interact with ALYREF.

CBC-ALYREF and 5' cap dependent mRNP export

ALYREF recruits the mRNP export machinery TREX complex to the 5' end of mRNA through direct interactions with both the CBC and TREX^{8,10}. The UBMs of ALYREF directly interact with the DDX39B component of the TREX complex¹⁰. The N-terminal UBM is included in the ALYREF construct used for our cryo-EM studies but did not show visible electron density. Thus, this UBM is likely exposed and available to interact with DDX39B, which further connects to the entire TREX complex (**Figure 5A**). Consistently, ALYREF, DDX39B, THOC1 and THOC2 are present in NCBP1 immunoprecipitations from RNase-treated HeLa cell nuclear extracts⁸. In yeast, mutually exclusive interactions were shown between Yra1 with Sub2 and the NXF1-NXT1 ortholog Mex67-Mtr2¹⁷. So, the ALYREF dependent NXF1-NXT1 loading on mRNA likely occurs after DDX39B dissociates from ALYREF. The CBC could also function as a landing pad for ALYREF as previously proposed³⁹. After recruitment to the 5' end of mRNA by the CBC, ALYREF could then transfer

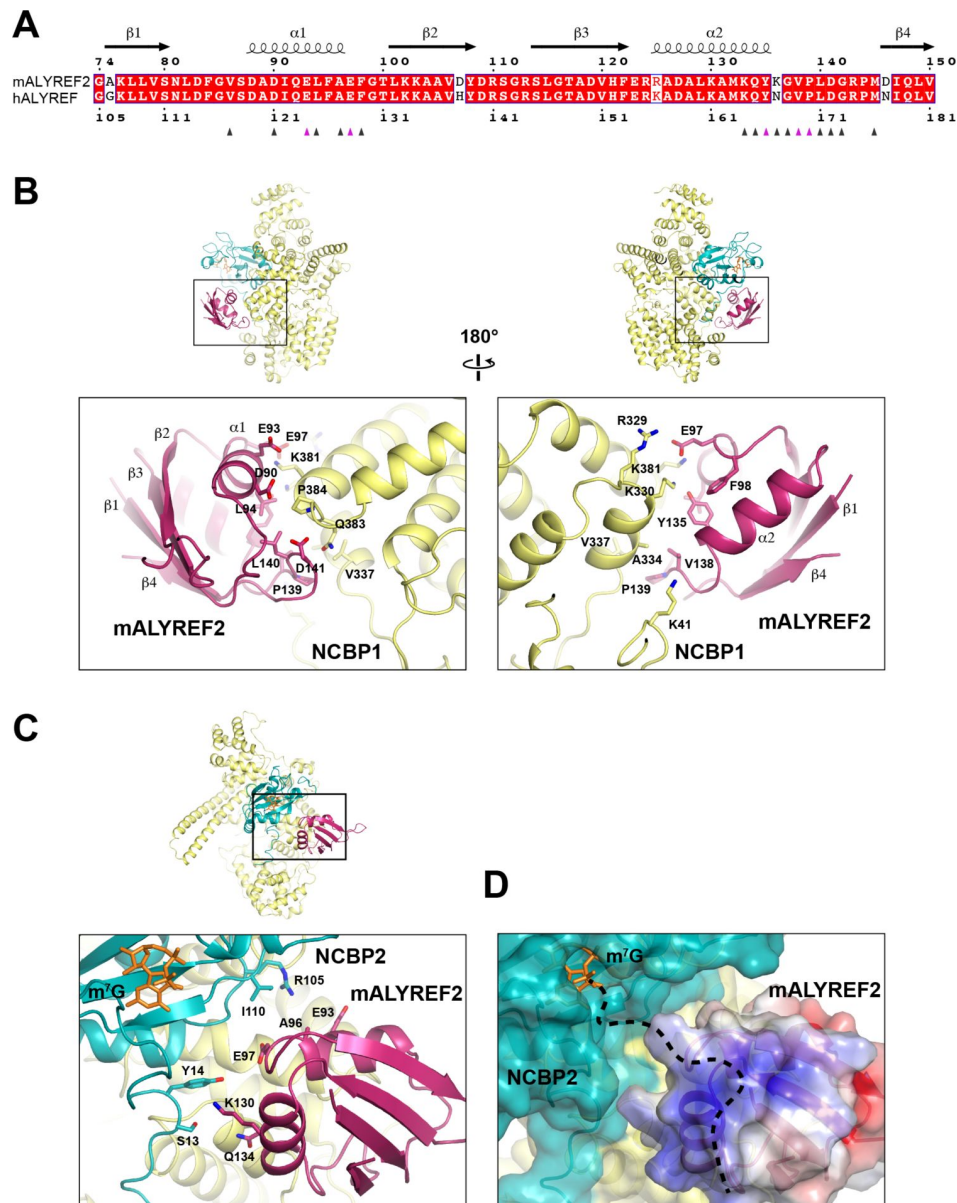


Figure 3.

ALYREF binds to both NCBP1 and NCBP2.

(A) Sequence alignment of the RRM domain of mALYREF2 and ALYREF⁶². The residues that interface with the CBC are indicated by triangles below the sequence. The triangles colored in purple correspond to residues subjected to mutagenesis in [Figure 4](#). (B) Details of the interaction between NCBP1 and mALYREF2. As a reference, an overall model is shown on the top to indicate the zoomed-in area. (C) Details of the interaction between NCBP2 and mALYREF2. (D) mALYREF2 features a positively charged surface near the cap analog bound to NCBP2. Surface of mALYREF2 is colored according to the electrostatic potential, ranging from red ($-5 k_B T/e$) to blue ($+5 k_B T/e$). Dotted line indicates putative RNA binding path.

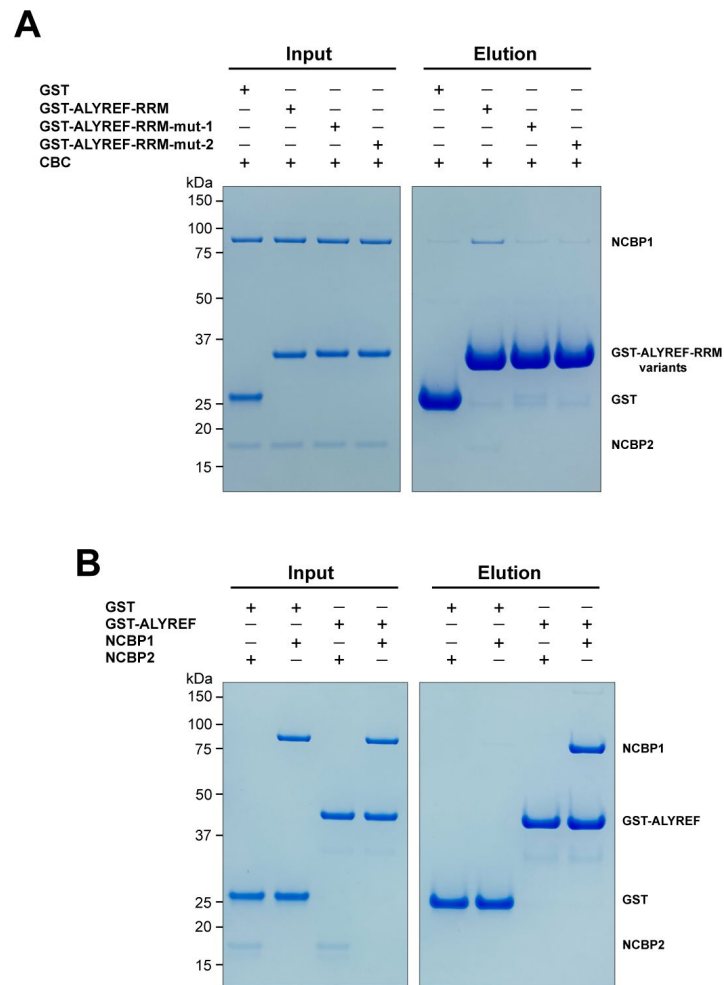


Figure 4.

Dissection of the ALYREF and CBC interfaces.

(A) Mutations of key interface residues on ALYREF reduced CBC binding. In vitro GST pull down assays were performed with purified recombinant human CBC and GST-tagged ALYREF-RRM wild type or mutants (mut-1, Y166R/V169R/P170R; mut-2, E124R/E128R). (B) NCBP1 is sufficient to interact with ALYREF. In vitro GST pull down assays were performed with purified recombinant GST-tagged ALYREF and individual CBC subunits. Results are representative of three independent experiments.

away from the 5' end, to other sites enriched with export factors and participate in different complexes located along the mRNA. In addition to the ALYREF-NXF1-NXT1 complex, some other ALYREF containing complexes could exist on the same mRNP, such as the complex of ALYREF/DDX39B/SARNP, which facilitates high order mRNP assembly^{40,41}.

The process of mRNP export, with the 5' end exiting first from the NPC, has been shown in both insect and human systems using electron microscopy and single molecule imaging techniques^{6,7}. Interestingly, in the latter study, several adjacent NPCs were found to engage in the export of the same mRNA⁴². This observation is reminiscent of the gene gating hypothesis which suggested that transcriptionally active genes are physically tethered to the site of mRNA export at the NPC⁴². Gated genes have been shown in yeast, worms, flies and humans^{43,44}. For these gated genes, the 5' directionality of mRNA export could be primarily driven by the key placement of crucial RNA export factors at the 5' end of the gene as illustrated here (**Figure 5A**), and this localization of export factors could greatly increase the efficiency of co-transcriptional processing and export.

CBC-ALYREF and viral hijacking of host mRNA export pathway

HSV-1 ICP27 and HVS ORF57 hijack the host mRNA export pathway through interactions with ALYREF. The RRM domain of ALYREF is targeted by both HSV-1 ICP27 and HVS ORF57 with overlapping interfaces (**Figure 5B**, **Figure 5-figure supplement 1**). Structural comparison between CBC-ALYREF, ALYREF-ICP27, and ALYREF-ORF57 reveals that the interface between ALYREF's RRM domain with the CBC is not compatible with the ICP27/ORF57-ALYREF interactions (**Figure 5B**, **Figure 5-figure supplement 1**). However, in vivo data shows that the ORF57 orthologue from Kaposi's Sarcoma-Associated Herpesvirus (KSHV) can still form a complex with ALYREF and the CBC²². So, it is likely that although ALYREF's RRM domain interface with the CBC could be disrupted by ORF57, ALYREF can still use the WxHD motif to interact with the CBC (**Figure 5C**). Using this strategy, a virus can hijack the host pathway while at the same time also disrupt host interactions and processes. It should also be noted that the CBC, ALYREF, ICP27, and ORF57 are all RNA binding proteins. In addition to the protein mediated interactions discussed above, RNA interactions should be considered, especially under the in vivo setting (**Figure 5C**). NXF1-NXT1 and DDX39B, the cellular ALYREF interacting proteins, are also hijacked by other factors from viruses. NXF1-NXT1 is targeted by influenza A virus NS1 protein⁴⁵ and SARS-CoV-2 Nsp1 protein⁴⁶. DDX39B is targeted by influenza A virus NP protein^{47,48}. The molecular mechanisms revealed here and from previous studies pave the way for new useful targets in anti-viral therapeutics.

Functional interplay of CBC-ALYREF and mRNP export factors

Transcription, splicing, and export are all tightly linked processes. The CBC promotes splicing through its binding partners, such as SRSF1^{49,50}. The detailed molecular interaction between the CBC and SRSF1 is revealed in a human pre-B^{act-1} spliceosome structure²⁸. The NCBP2 subunit of the CBC is the major binding site for SRSF1. Structural overlay of this structure with the CBC-ALYREF structure shows no significant steric hindrance between SRSF1 and ALYREF (**Figure 6A**). However, whether the CBC-ALYREF-SRSF1 complex exits in vivo and how their functions might be coordinated require further studies. After splicing, SRSF1 is mainly deposited on exons^{51,52}. SRSF1 functions in mRNA export through interaction with the export receptor NXF1-NXT1^{26,53,54}. Of note, the interaction between both ALYREF and SR proteins with NXF1-NXT1 is regulated by phosphorylation. Only hypo-phosphorylated SR proteins can bind to NXF1-NXT1 efficiently^{26,55,56}. Interestingly, SR proteins, such as Gbp2 and Hrb1 in *S. cerevisiae*, also interact directly with the TREX complex^{11,57}, suggesting coordinated actions of TREX and SR proteins in mRNA export.

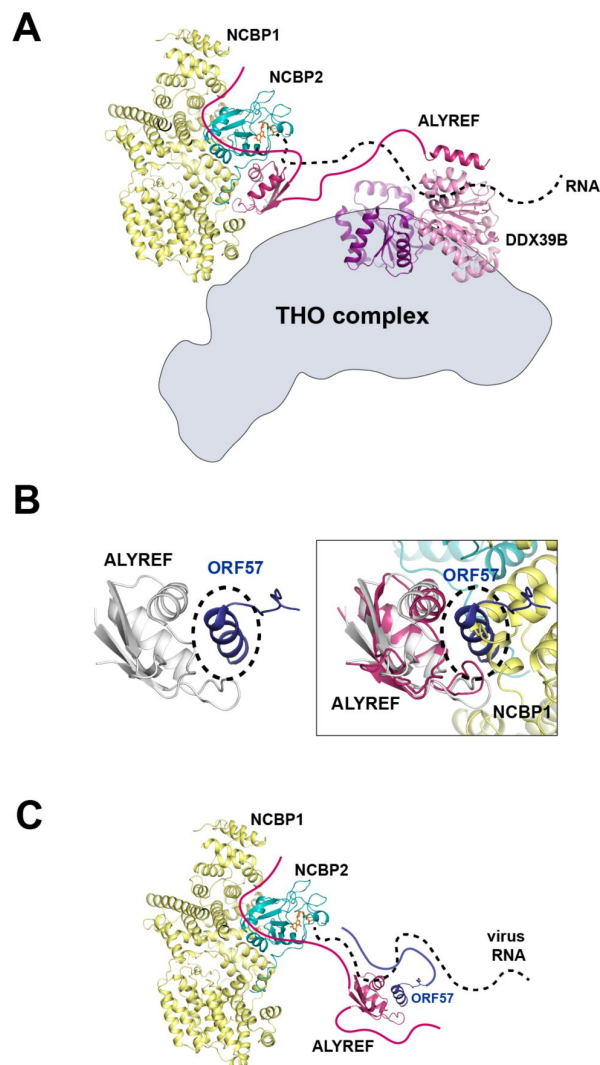


Figure 5.

Functional implications for ALYREF and CBC in 5' cap dependent mRNP export.

(A) ALYREF links the TREX complex to the CBC. The RRM domain of ALYREF recognizes the CBC at the 5' end of mRNA. The UBM of ALYREF binds to the DDX39B subunit of TREX; their complex is represented using their yeast orthologs Yra1 and Sub2, respectively (PDB ID 5SUP). DDX39B in turn associates with the THO subcomplex of TREX. (B) HVS ORF57 binds to the RRM domain of ALYREF and interferes with the CBC-ALYREF interaction. (left) NMR structure of the ALYREF-ORF57 complex (PDB ID 2YKA). (right) The CBC-ALYREF structure is overlaid with the ALYREF-ORF57 structure. (C) Model of viral mRNA export mediated by herpes viral ORF57 homologs. On viral transcripts, ALYREF associates with the CBC and ORF57 via the WxHD motif and the RRM domain, respectively. Both ALYREF and ORF57 feature RNA binding regions to form contacts with the RNA. ALYREF recruits the other TREX complex components to facilitate nuclear export of viral mRNAs.

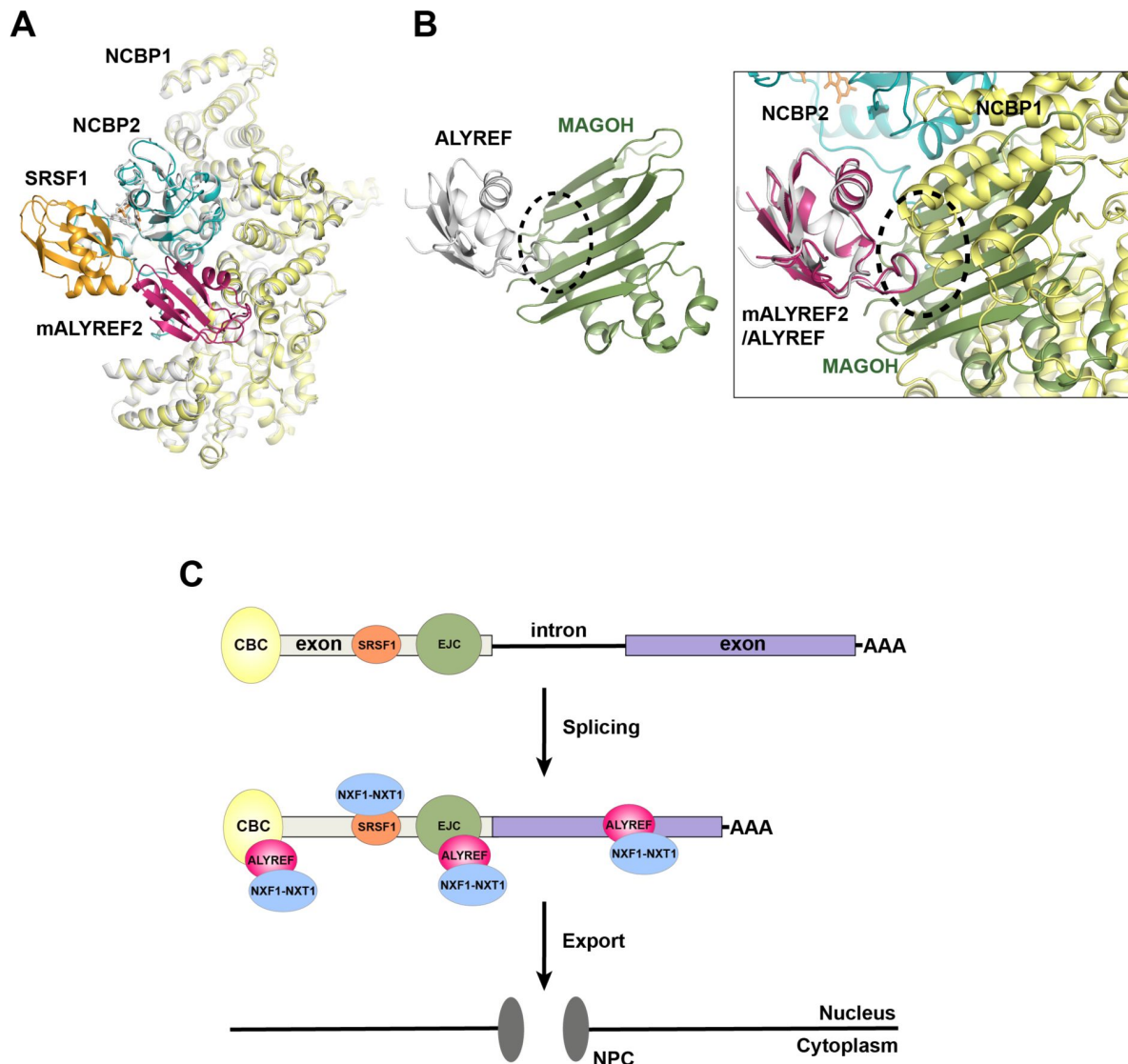


Figure 6.

ALYREF and CBC in splicing and export

(A) Overlay of the CBC-ALYREF structure with the CBC-SRSF1 structure (PDB ID 7ABG). CBC-ALYREF is colored as in **Figure 2B**. CBC-SRSF1 is colored with the CBC in white and SRSF1 in orange. (B) ALYREF interaction with the CBC and the EJC is mutually exclusive. (left) ALYREF binds to the MAGOH subunit of the EJC (PDB ID 7ZNI). (right) The CBC-ALYREF structure is overlaid with the ALYREF-MAGOH structure. (C) Model of the mRNP export receptor NXF1-NXT1 recruitment by CBC-ALYREF and other factors during mRNA maturation.

Unlike the CBC and SRSF1 interfaces discussed above, the interaction between the ALYREF RRM domain and the CBC is not compatible with the interface between the ALYREF RRM domain and the EJC subunit MAGOH¹² (Figure 6B). Mutation of the ALYREF WxHD motif affects its interaction with both the CBC and the EJC subunit eIF4A3³². Since both the WxHD motif and the RRM domain of ALYREF are mutually exclusive binding sites for the CBC and the EJC, the formation of the EJC-ALYREF complex likely happens after ALYREF dissociates from the CBC. It is also possible that the EJC-ALYREF interaction is independent of the CBC (Figure 6C). This possibility is supported by the report that ALYREF can be recruited to RNA by both CBC dependent and independent mechanisms⁵⁸. The resulting mRNP with multiple copies of ALYREF and NXF1-NXT1, each recruited through different mechanisms and at different sites on the mRNP (Figure 6C), could exhibit increased export efficiency.

Materials and methods

Plasmids

Human NCBP1 (UniProt Q09161) was cloned into the pFastBac HTc vector with an N-terminal TEV cleavable His tag. Human NCBP2 (UniProt P52298) was cloned into a modified pFastBac1 vector containing an N-terminal TEV cleavable GST tag. Human ALYREF (UniProt Q86V81) and mouse ALYREF2 (UniProt Q4KL64) constructs were cloned into a modified pGST-4T-1 vector containing an N-terminal TEV cleavable GST tag.

Protein expression and purification

The NCBP1-NCBP2 complex was expressed in High-Five insect cells (Invitrogen) by coinfection of recombinant baculoviruses. Individual NCBP1 and NCBP2 subunits were expressed in High-Five insect cells infected with the respective recombinant baculovirus. High-Five cells were harvested 48 hrs after infection and lysed in a buffer containing 50 mM Tris pH 8.0, 300 mM NaCl, 0.2 mM AEBSF, 2 mg/L aprotinin, 1 mg/L pepstatin, 1 mg/L leupeptin, and 0.5 mM TCEP. Proteins were purified using Glutathione Sepharose 4B resin (Cytiva) for NCBP2 and NCBP1-NCBP2, or Ni Sepharose 6FF resin (Cytiva) for NCBP1 alone. The proteins were further purified on a mono Q column (Cytiva). The expression tags were removed by overnight incubation with GST-TEV (for NCBP2 and NCBP1-NCBP2) or His-TEV (for NCBP1) at 4 °C. The samples were passed through Glutathione Sepharose 4B resin (for NCBP2 and NCBP1-NCBP2) or Ni Sepharose 6FF resin (for NCBP1) to remove undigested protein and TEV. The proteins were further purified on a Superdex 200 column (Cytiva) in 10 mM Tris pH 8.0, 300 mM NaCl, and 0.5 mM TCEP.

GST tagged ALYREF (residues 1-183), ALYREF-RRM (residues 102-183), ALYREF-RRM-mut-1 (Y166R/V169R/P170R), ALYREF-RRM-mut-2 (E124R/E128R), and mALYREF2 (residues 1-155) were expressed in *E. coli* Rosetta cells (Sigma-Aldrich). Protein expression was induced with 0.5 mM IPTG at 16°C overnight. Cells were lysed in a buffer containing 50 mM Tris, pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM arginine, 0.5 mM TCEP, 0.2 mM AEBSF, and 2 mg/L aprotinin. The GST-tagged proteins were pulled down using Glutathione Sepharose 4B resin. GST-tagged ALYREF-RRM wild type and mutant proteins were purified on a Superdex 200 column in 10 mM Tris pH 8.0, 300 mM NaCl, and 5 mM DTT. GST-tagged ALYREF (residues 1-183) was purified by a cation exchange column (source 15S, Cytiva), followed by a Superdex 200 column equilibrated with 10 mM Tris, pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM Arginine, and 0.5 mM TCEP. GST-tagged mALYREF2 (residues 1-155) was purified by an anion exchange column (Q Sepharose, Cytiva), followed by a cation exchange column (SP Sepharose, Cytiva). For untagged mALYREF2 used in cryo-EM studies, the GST tag was removed by overnight incubation with GST-TEV at 4 °C. Untagged mALYREF2 was further purified on a cation exchange HiTrap SP column (Cytiva), followed by a Superdex 200 column equilibrated with 10 mM Tris, pH 8.0, 150 mM NaCl, 50 mM glutamic acid, 50 mM Arginine, and 0.5 mM TCEP.

All purified proteins were flash frozen in liquid nitrogen, and stored at -80°C .

Cryo-EM sample preparation and data collection

NCBP1-NCBP2 at $1.2\ \mu\text{M}$ was incubated with mALYREF2 (residues 1-155) at $3.6\ \mu\text{M}$ in the presence of the 5' cap analog m^7GpppG (NEB) at $500\ \mu\text{M}$ at 4°C for 0.5 hr. The sample was deposited on glow-discharged UltrAuFoil R 1.2/1.3 grids (Quantifoil). Grids were blotted for 6 s with a blotting force of 6 at 4°C and 100% humidity and plunged into liquid ethane using a FEI VitroBot Mark IV (Thermo Fisher). The data were collected with a Glacios Cryo-TEM (Thermo Fisher) equipped with a Falcon 4i detector (Thermo Fisher).

Movies were collected with EPU at a magnification of 190,000x, corresponding to a calibrated pixel size of $0.732\ \text{\AA}/\text{pixel}$. A total of 5858 movies recorded in EER format were collected with a defocus range from $1.0\ \mu\text{m}$ to $2.0\ \mu\text{m}$. A full description of the cryo-EM data collection parameters can be found in [Table 1](#).

Cryo-EM image processing and Model building

Cryo-EM data were processed with cryoSPARC⁵⁹. Movies in EER format were gain normalized, aligned and dose-weighted using patch motion correction, followed by patch CTF estimation. About 2.5 million particles were initially picked with templates generated from the NCBP1-NCBP2 structure (PDB 1H6K). Multiple rounds of 2D classification and heterogenous refinement were carried out to obtain 745k good particles. Refinement of these particles showed unambiguous density corresponding to mALYREF2. To select mALYREF2 containing particles, heterogenous refinement was performed with two classes corresponding to NCBP1-NCBP2-mALYREF2 and NCBP1-NCBP2 alone. The class corresponding to NCBP1-NCBP2 was subjected to another round of heterogenous refinement to get a cleaner set of particles, followed by homogenous refinement and local refinement. The NCBP1-NCBP2 map was refined with 143k particles and achieved a resolution of $3.51\ \text{\AA}$ as assessed by an FSC threshold of 0.143. The class corresponding to NCBP1-NCBP2-mALYREF2 was subjected to homogenous refinement and local refinement. The NCBP1-NCBP2-mALYREF2 map was refined with 440k particles and achieved a resolution of $3.16\ \text{\AA}$ as assessed by an FSC threshold of 0.143.

An initial model of NCBP1-NCBP2 was obtained by docking AlphaFold models of NCBP1 (AF-Q09161-F1) and NCBP2 (AF-P52298-F1) into the cryo-EM density map sharpened using a B-factor of $-121\ \text{\AA}^2$ estimated from the Guinier plot. An initial model of NCBP1-NCBP2-mALYREF2 was obtained by docking AlphaFold models of NCBP1, NCBP2, and mALYREF2 into the cryo-EM density map sharpened using a B-factor of $-130\ \text{\AA}^2$ estimated from the Guinier plot. The models were adjusted in Coot⁶⁰, followed by real-space refinement in Phenix⁶¹. The final NCBP1-NCBP2 model contains NCBP1, NCBP2, and the m^7G moiety of m^7GpppG . The final NCBP1-NCBP2-mALYREF2 model contains NCBP1, NCBP2, the RRM domain of mALYREF2, and the m^7G moiety of m^7GpppG . Figures were prepared using PyMOL (Molecular Graphics System, Schrodinger, LLC) and Chimera¹⁵.

GST pull down

GST or GST tagged proteins were pre-incubated with GST resin in binding buffer (10 mM Tris pH 8.0, 500 mM NaCl, 50 mM glutamic acid, 50 mM arginine, 0.5 mM TCEP) on ice for 0.5 hour and was mixed with gentle tapping every 3-5 minutes. The beads were then washed three times with 600 μl of buffer containing 10 mM Tris pH 8.0, 100 mM NaCl (**Figure 1B** and **4B**) or 50 mM NaCl (**Figure 4A**), and 0.5 mM TCEP. NCBP1, NCPB2, or the NCBP1-NCBP2 complex was incubated with the 5' cap analog m^7GpppG at $20\ \mu\text{M}$, adjusted salt concentration to 100 mM NaCl (**Figure 1B** and **4B**) or 50 mM NaCl (**Figure 4A**), and then added to beads. The samples were incubated on ice for 0.5 hour and mixed with gentle tapping every 3-5 minutes. Beads were then washed three times with 600 μl of wash buffer (10 mM Tris pH 8.0, 50 mM NaCl, 0.5 mM TCEP) before bound

proteins were eluted in 10 mM Tris pH 8.0, 500 mM NaCl (**Figure 1B** [↗](#) and **4B** [↗](#)) or 150 mM NaCl (**Figure 4A** [↗](#)), 25 mM Glutathione, and 0.5 mM TCEP. 6% (**Figure 1B** [↗](#) and **4B** [↗](#)) or 3% (**Figure 4A** [↗](#)) of the input and 60% of the eluted proteins were analyzed using Coomassie stained SDS-PAGE gels. The experiments were repeated three times independently.

Data availability

The cryo-EM models and corresponding maps have been deposited in the RCSB database and in the EMDB.

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Conflict of interest

The authors declare they have no conflict of interest.

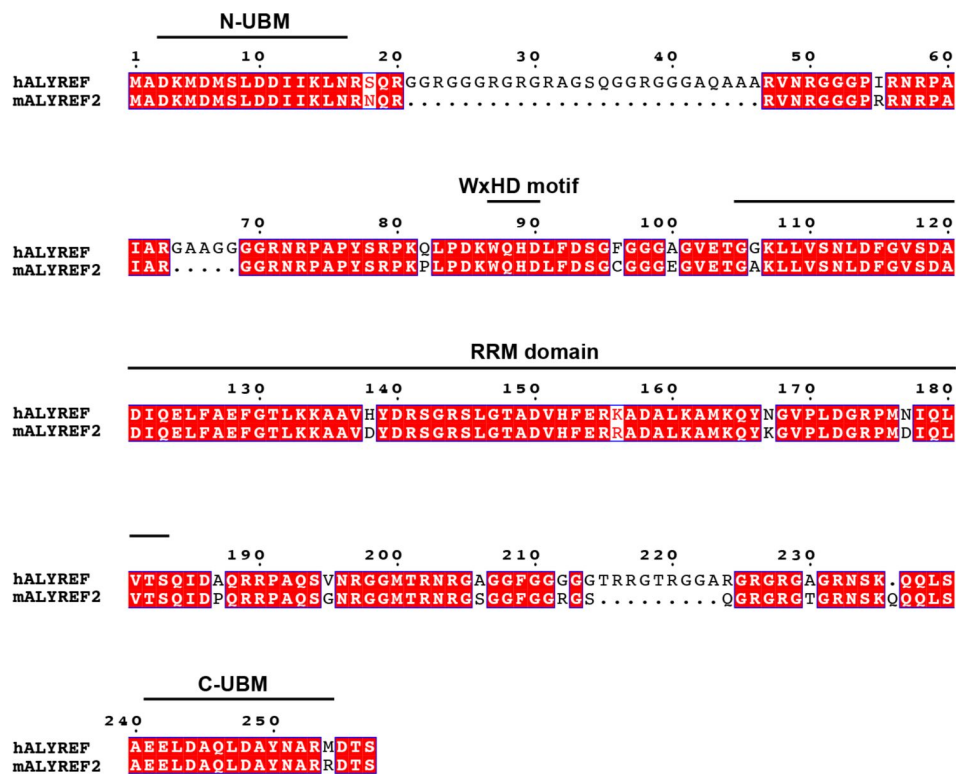


Figure 1 – figure supplement 1.

Human ALYREF and mouse ALYREF2 are conserved.

Sequence alignment of ALYREF and mALYREF2⁶². The DDX39B binding motifs (N-UBM and C-UBM), the WxHD motif, and the RRM domain are indicated.

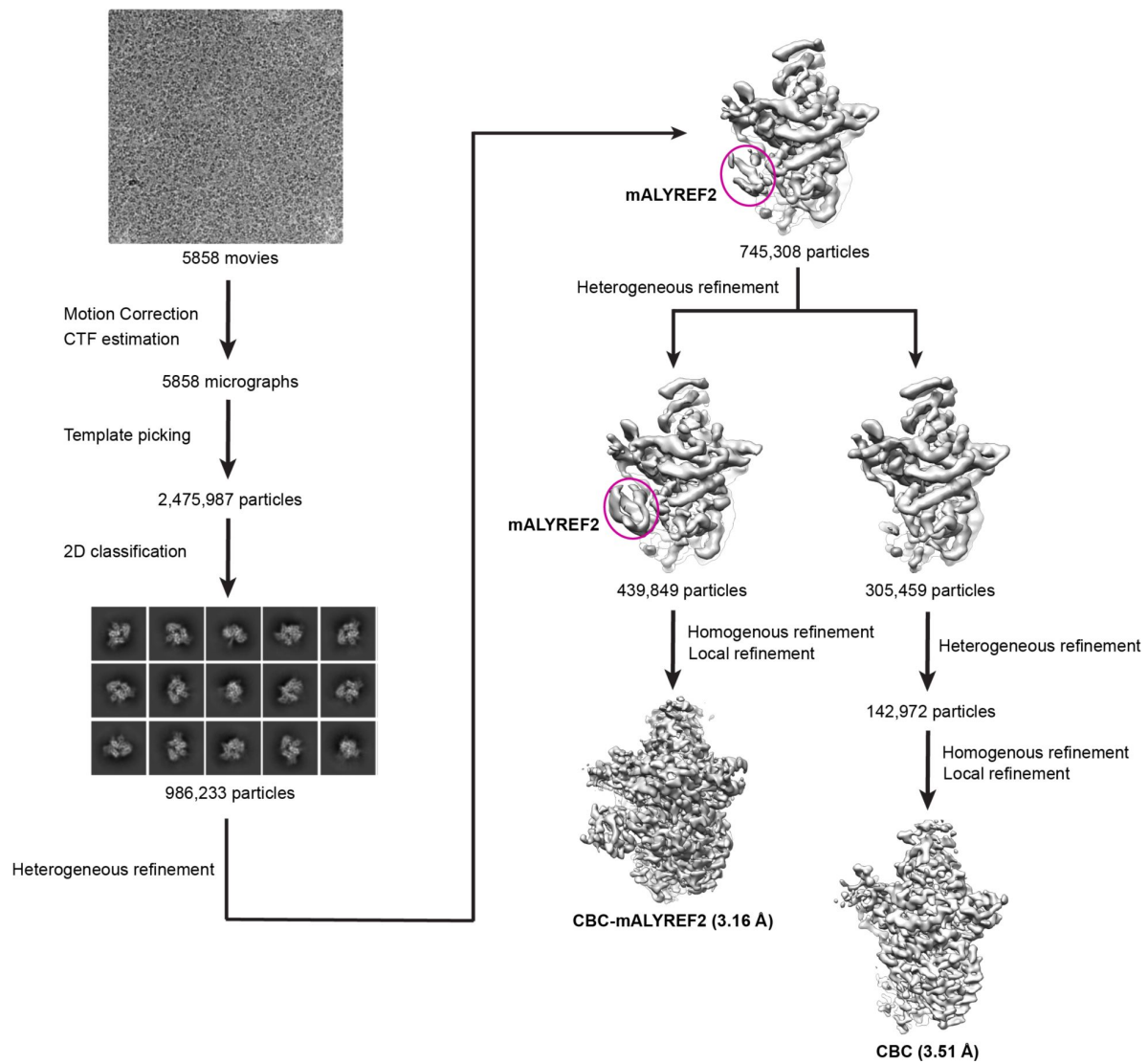


Figure 2 – figure supplement 1.

Workflow for cryo-EM data processing.

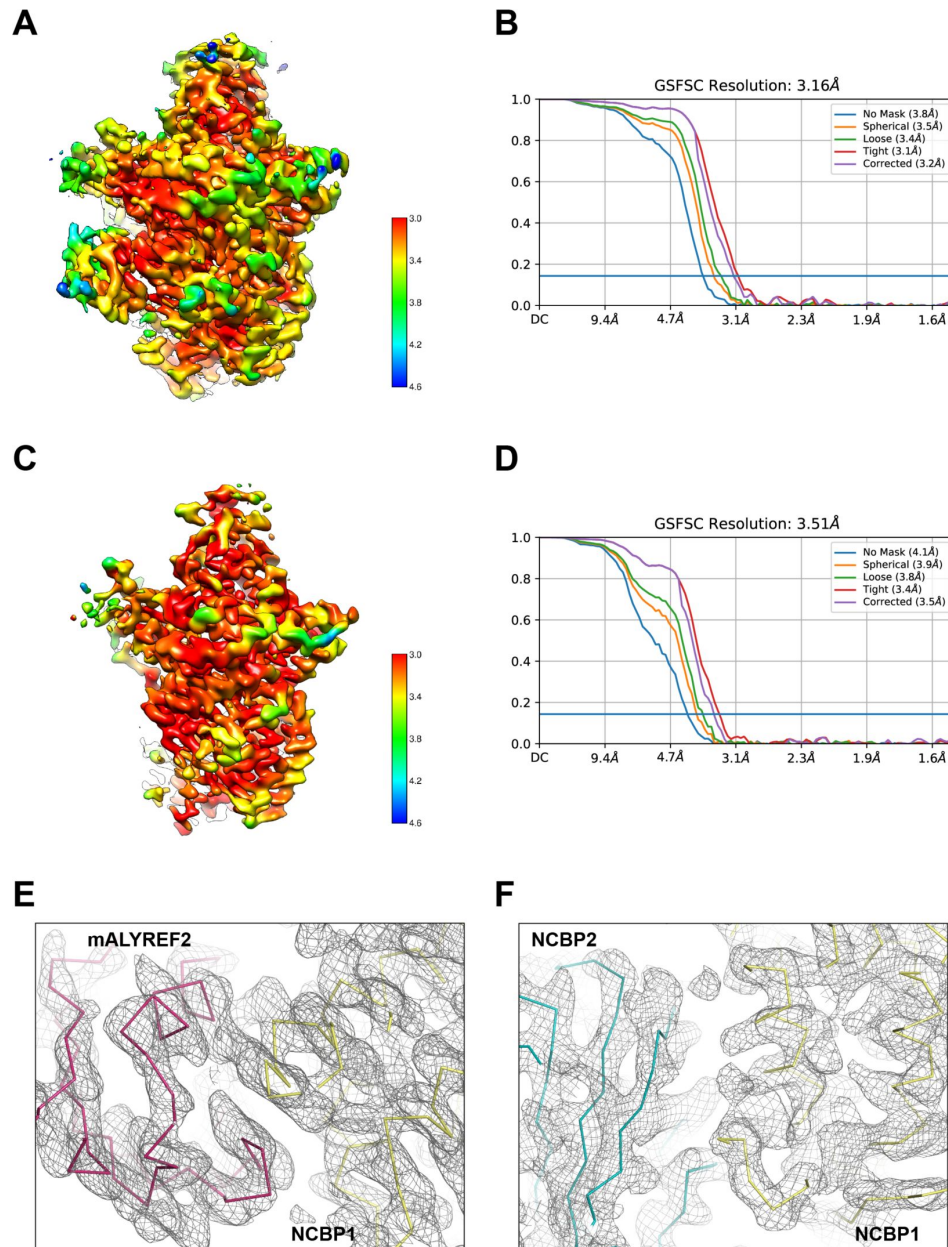


Figure 2 – figure supplement 2.

Local resolution estimation and FSC curves.

(A) Local resolution of the CBC density map. (B) FSC curves between the two half maps of CBC. (C) Local resolution of the CBC-mALYREF2 density map. (D) FSC curves between the two half maps of CBC-mALYREF2. (E) Electron density map of CBC-mALYREF2 at the interface between NCBP1 and mALYREF2. (F) Electron density map of CBC-mALYREF2 at the interface between NCBP1 and NCBP2.

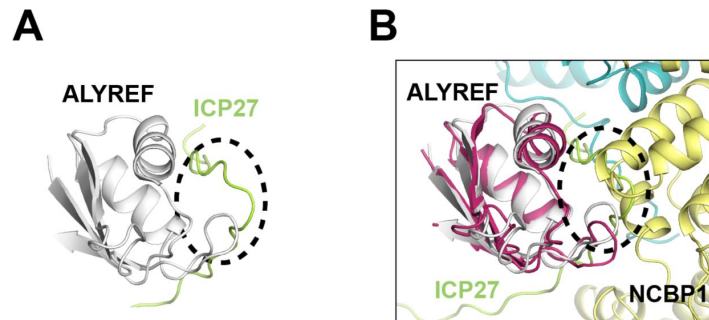


Figure 5 – figure supplement 1.

HSV-1 ICP27 binds to the RRM domain of ALYREF and interferes with the CBC-ALYREF interaction.

(A) NMR structure of the ALYREF-ICP27 complex (PDB ID 2KT5). (B) The CBC-ALYREF structure is overlaid with the ALYREF-ICP27 structure.

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Editors

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Reviewer #1 (Public Review):

Summary:

The authors use a combination of biochemistry and cryo-EM studies to explore a complex between the cap-binding complex and an RNA binding protein, ALYREF, that coordinates mRNA processing and export.

Strengths:

The biochemistry and structural biology are supported by mutagenesis which tests the model in vitro. The structure provides new insight into how key events in RNA processing and export are likely to be coordinated.

Weaknesses:

The authors provide biochemical studies to confirm the interactions that they identify; however, they do not perform any studies to test these models in cells or explore the consequences of mRNA export from the nucleus. In fact, several of the amino acids that they identified in ALYREF that are critical for the interaction, as determined by their own biochemical studies, are conserved in budding yeast Yra1 (residues E124/E128 are E/Q in budding yeast and residues Y135/V138/P139 are F/S/P), where the impact on poly(A) RNA export from the nucleus could be readily evaluated. The authors could at least mention this point as part of the implications and the need for future studies. No one seems to have yet targeted any of these conserved residues, so this would be a logical extension of the current work.

Specific suggestions:

The authors could put their work in context by speculating how some of the amino acids that they identify as being critical for the interactions they identify could contribute to cancer. For example, they mention mutations of interacting residues in NCBP2 are associated with human cancers, pointing out that NCBP2 R105C amino acid substitution has been reported in colorectal cancer and the NCBP2 I110M mutation has been found in head and neck cancer. Do the authors speculate that these changes would decrease the interaction between NCBP2 and ALYREF and, if so, how would this contribute to cancer? They also mention that a K330N mutation in NCBP1 in human uterine corpus endometrial carcinoma, where Y135 on the $\alpha 2$ helix of mALYREF2 makes a hydrogen bond with K330 of NCBP1. How do they speculate loss of this interaction would contribute to cancer?

Reviewer #2 (Public Review):**Summary:**

In this manuscript, Bradley and his colleagues represented the cryo-EM structure of the nuclear cap-binding complex (CBC) in complex with an mRNA export factor, ALYREF, providing a structural basis for understanding CBC regulating gene expression.

Strengths:

The authors successfully modeled the N-terminal region and the RRM domain of ALYREF (residues 1-183) within the CBC-ALYREF structure, which revealed that both the NCBP1 and NCBP2 subunits of the CBC interact with the RBM domain of ALYREF. Further mutagenesis and pull-down studies provided additional evidence to the observed CBC-ALYREF interface. Additionally, the authors engaged in a comprehensive discussion regarding other cellular complexes containing CBC and/or ALYREF components. They proposed potential models that elucidated coordinated events during mRNA maturation. This study provided good evidence to show how CBC effectively recruits mRNA export factor machinery, enhancing our understanding of CBC regulating gene expression during mRNA transcription, splicing, and export.

Weaknesses:

No in vivo or in vitro functional data to validate and support the structural observations and the proposed models in this study. Cryo-EM data processing and structural representation need to be strengthened.

Reviewer #3 (Public Review):**Summary:**

The authors carried out structural and biochemical studies to investigate the multiple functions of CBC and ALYREF in RNA metabolism.

Strengths:

For the structural study part, the authors successfully revealed how NCBP1 and NCBP2 subunits interact with mALYREF (residues 1-155). Their binding interface was then confirmed by biochemical assays (mutagenesis and pull-down assays) presented in this study.

Weaknesses:

The authors did not provide functional data to support their proposed models. The authors should include more details regarding the workflow of their cryo-EM data processing in the figure.